

Mahrokh Ghoddousi Boroujeni



Portfolio

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About me

I am a PhD candidate at École Polytechnique Fédérale de Lausanne (EPFL), Switzerland, working at the intersection of machine learning and control. My research focuses on learning models and controllers from limited data that ensure stability and reliable performance when deployed in real-world scenarios under unseen conditions. I have experience in rigorous problem formulation, principled algorithm design, and empirical evaluation of the developed methods.

Originally from Tehran, Iran, I am currently based in Lausanne, Switzerland, and hold a type B residence permit.

Work Experience

Teaching Assistant, EPFL

Assisted in courses Multivariable Control, Networked Control Systems, and Control Systems. Responsibilities included designing assignments and projects, leading exercise sessions, providing hands-on guidance to students, and grading exams.

Lausanne, Switzerland

Sept. 2021 – Nov. 2025

Student Supervision, EPFL

Supervised 10 semester-long research projects: 4 in Robotics and Reinforcement Learning, and 6 in Machine Learning for energy systems. Initiated project proposals, provided technical mentorship, and guided students through all phases of research and implementation.

Lausanne, Switzerland

Sept. 2021 – Nov. 2025

Research Intern, Max Planck Institute for Intelligent Systems

Developed a framework for walking and running of biped robots, relaxing the common assumption of a fixed center of mass height to enable diverse gaits with minimal added complexity. Derived theoretical results and contributed to hardware deployment ([publication](#)).

Tübingen, Germany

March 2020 – Sept. 2020

Research Intern, EPFL

Developed a software for energy flow optimization between photovoltaic cells, electric vehicles, batteries, and buildings. The tool incorporates battery degradation models and real-world operational constraints to improve practicality.

Neuchâtel, Switzerland

June 2019 – Sept. 2019

Research Intern, ETH Zürich

Designed a privacy-preserving adaptive pricing mechanism for smart grids, addressing household data privacy in real-time electricity markets ([publication](#)).

Zürich, Switzerland

July 2018 – Sept. 2018

Mathematics Olympiad Teacher, Farzanegan High School

Prepared high school students for Iran's National Mathematics Olympiad through problem-solving training.

Tehran, Iran

Sept. 2015 – Sept. 2016

Volunteer Teacher, Local Orphanage

Provided academic support for underserved youth preparing for university entrance. Gained experience in inclusive education, adapting to the needs of students with learning difficulties.

Tehran, Iran

Sept. 2015 – July 2016

Education

EPFL, PhD Candidate in the Robotics and Intelligent Systems Doctoral School

- Thesis: *Learning and Control with Generalization and Stability Certificates*
- Supervised by Prof. Giancarlo Ferrari-Trecate and Prof. Andreas Krause

Lausanne, Switzerland

Nov. 2020 – Nov. 2025

Sharif University of Technology, BSc in Electrical Engineering (Control specialization) and Computer Science

- Completed a dual-degree program

Tehran, Iran

Sep. 2015 – July 2020

- Thesis: *Camera-Based Real-Time Autonomous Racing Cars*
- GPA: 18.45/20; ranked top 1 in the Control group ([Transcript](#))

Honors and Awards

- Outstanding Student Paper Award**, *IEEE Conference on Decision and Control (CDC)*, 2024
 For the paper *A PAC-Bayesian Framework for Optimal Control with Stability Guarantees*. Awarded to 3 out of 1000 first-author students, based on the paper’s originality, clarity, and potential impact.
- E3 Fellowship**, EPFL, 2019
 Awarded as part of a competitive undergraduate research program (acceptance rate < 2%).
- Top 0.1% Nationwide**, *Iranian National University Entrance Exam*, 2015
 Ranked 164th out of over 182,000 participants.

Selected Publications

- M. G. Boroujeni**, C. L. Galimberti, A. Krause, and G. Ferrari-Trecate. A PAC-Bayesian Framework for Optimal Control with Stability Guarantees. *IEEE Conference on Decision and Control (CDC)*, 2024.
 Proposed a framework for designing data-driven neural controllers that ensures stability and generalization beyond the training data. Demonstrated the efficacy of the approach in simulation on a cooperative robotics task ([PDF](#)).
- A. Abyaneh, **M. G. Boroujeni**, H. Lin, and G. Ferrari-Trecate. Contractive Dynamical Imitation Policies for Efficient Out-of-Sample Recovery. *International Conference on Learning Representations (ICLR)*, 2025.
 Introduced an imitation learning framework that ensures reliable recovery from out-of-sample perturbations. Validated the method on robotic manipulation and navigation tasks in the Isaac Sim simulator ([PDF](#)).
- M. G. Boroujeni**, E. Daneshmand, L. Righetti, and M. Khadiv. A Unified Framework for Walking and Running of Bipedal Robots. *International Conference on Advanced Robotics (ICAR)*, 2021.
 Developed a low-complexity framework for generating diverse walking and running gaits in bipedal robots. Demonstrated robust performance in simulation with the Bolt robot under disturbances in the environment ([PDF](#)).

Skills

Machine Learning & AI	Imitation learning; meta & federated learning; generalization bounds; probabilistic inference (e.g., variational methods, MCMC) & modeling (e.g., Gaussian processes, Bayesian networks, Markov models); neural networks (recurrent, convolutional, adversarial); computer vision
Control & Systems	Learning-based control; stability guarantees; system identification; stochastic systems; networked control systems
Programming	Expert in Python, MATLAB, Java, R; familiar with C++, Bash, CodeVision
Simulation	Expert in MuJoCo, Isaac Sim, Simulink; familiar with ROS
Research & Communication	Scientific writing & presentation; student supervision & mentoring; cross-disciplinary collaboration (robotics, energy systems, photovoltaics)
Languages	Persian (native), English (fluent), French (conversational)